

Annotation Validation Report

Comparing Automated Topic Analysis Against Human Annotation

This report evaluates two automated topic classification methods — a keyword-based pipeline and a semantic (AI embedding) tagger — against human annotation across 400 labeled articles and 12 topics.

Annotator 1 and Annotator 3 each independently labeled a separate set of 200 articles from the Black media corpus.

400

Articles Evaluated

12

Topics

2

Human Annotators

2

Pipeline Methods

What We Are Comparing

Two automated pipelines were validated against hand-labeled articles.

Method 1 — Keyword Pipeline

Scans each article's title, description, and body text for a curated dictionary of seed phrases (e.g. 'police brutality', 'voter suppression', 'redlining'). If a phrase is found, the article is tagged with that topic.

Strengths: Fast, transparent, easy to audit.

Weaknesses: Misses articles that use different wording for the same concept.

Method 2 — Semantic Tagger

Uses an AI sentence-embedding model (all-MiniLM-L6-v2) to measure the meaning similarity between each article and each topic description. Tags the article if similarity exceeds a threshold.

Strengths: Captures meaning, not just exact words.

Weaknesses: Less transparent; threshold-sensitive.

Human Annotation Setup

Two trained annotators independently labeled separate sets of 200 articles each (400 total).

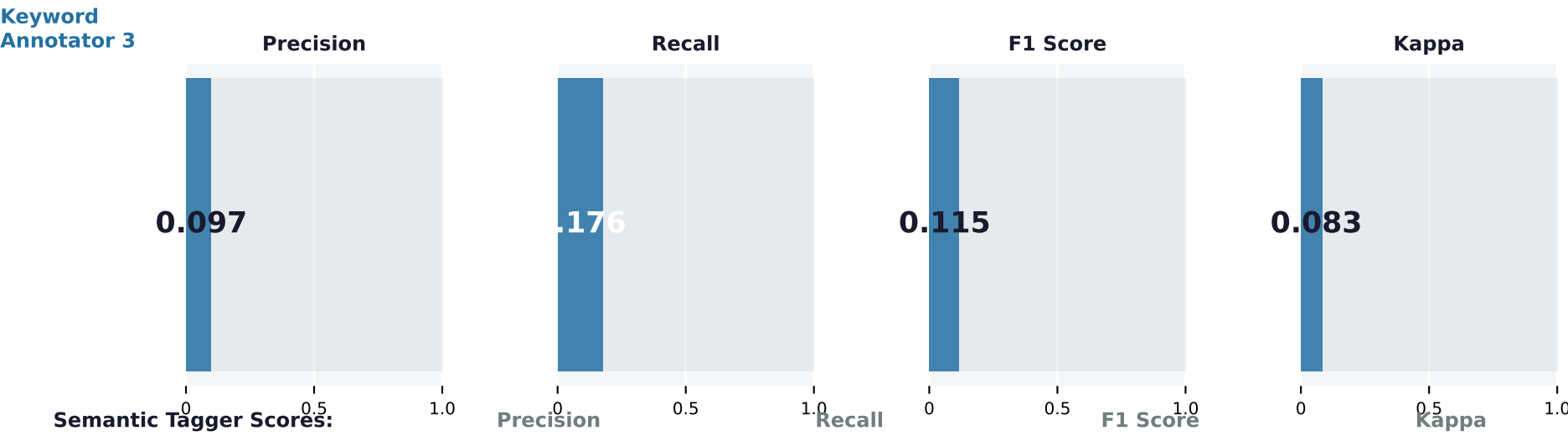
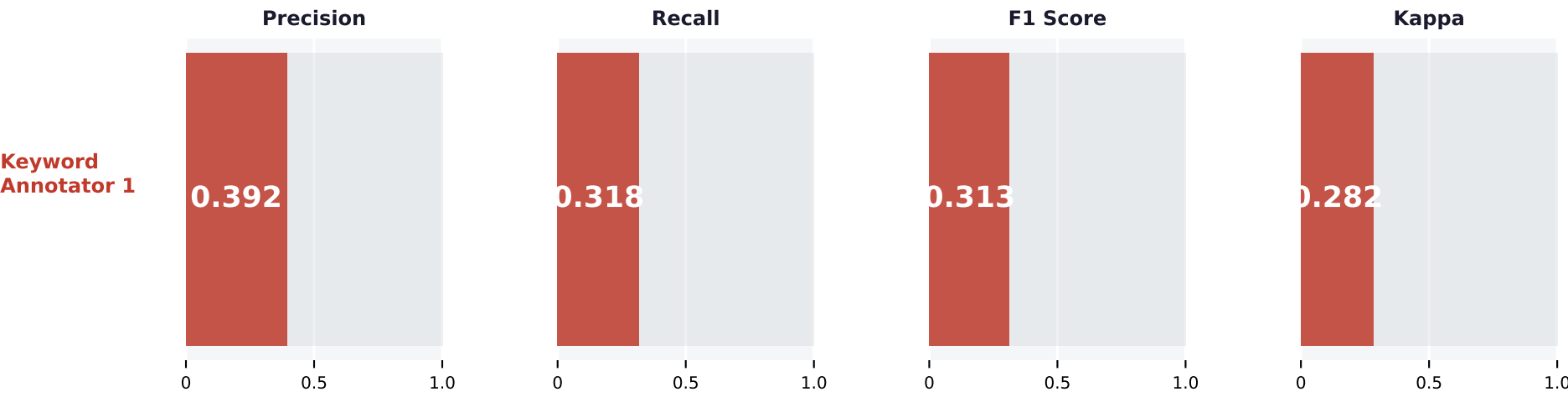
For each article they marked which of the 12 social conflict topics were present using a structured Excel sheet.

The human labels serve as the 'ground truth' against which both automated pipelines are evaluated.

Because the annotators labeled different articles, results are reported separately for each annotator's set.

Overall Performance Scorecard

Macro-averaged scores across all 12 topics for each method and annotator set.



Semantic Tagger Scores:

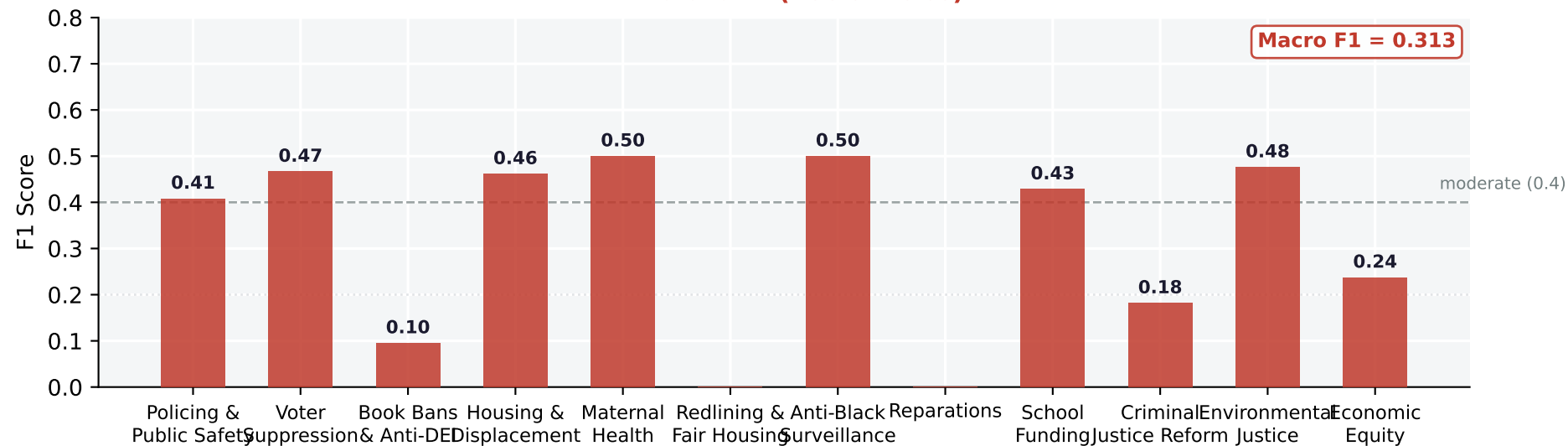
Semantic — Annotator 1	0.197	0.586	0.257	0.200
Semantic — Annotator 3	0.233	0.469	0.275	0.250

Precision: when the pipeline tags a topic, how often is it correct? Recall: of all articles a human labeled, how many did the pipeline catch?
F1: overall balance between precision and recall. Kappa: agreement corrected for chance — the most rigorous metric (≥ 0.4 = substantial, 0.2-0.4 = fair, < 0.2 = poor).

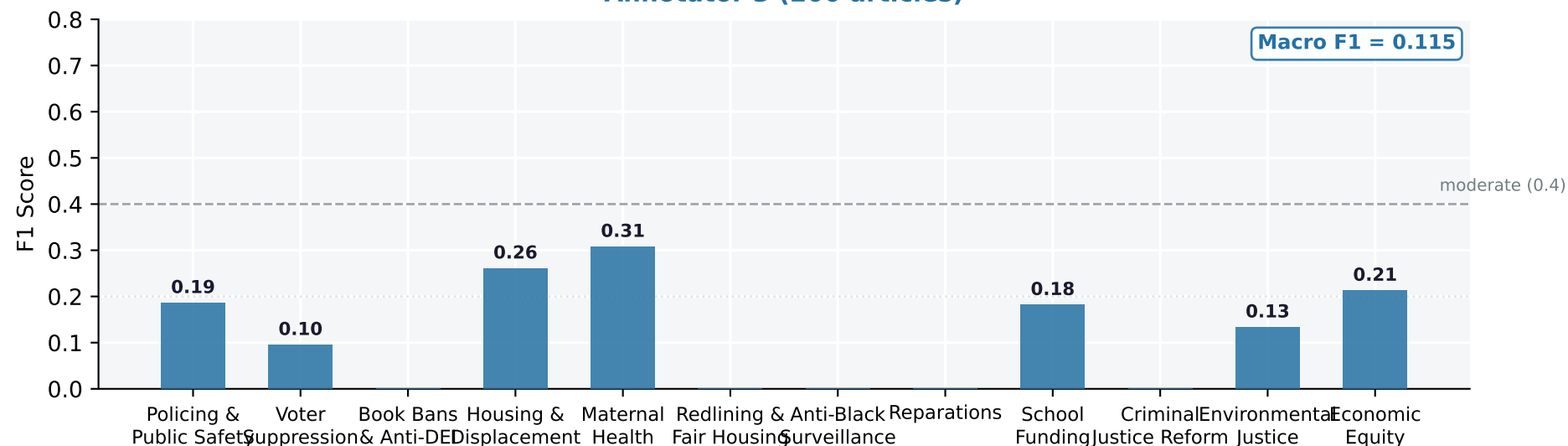
Keyword Pipeline — F1 Score by Topic

How well does the keyword pipeline agree with each annotator on each topic?

Annotator 1 (200 articles)



Annotator 3 (200 articles)

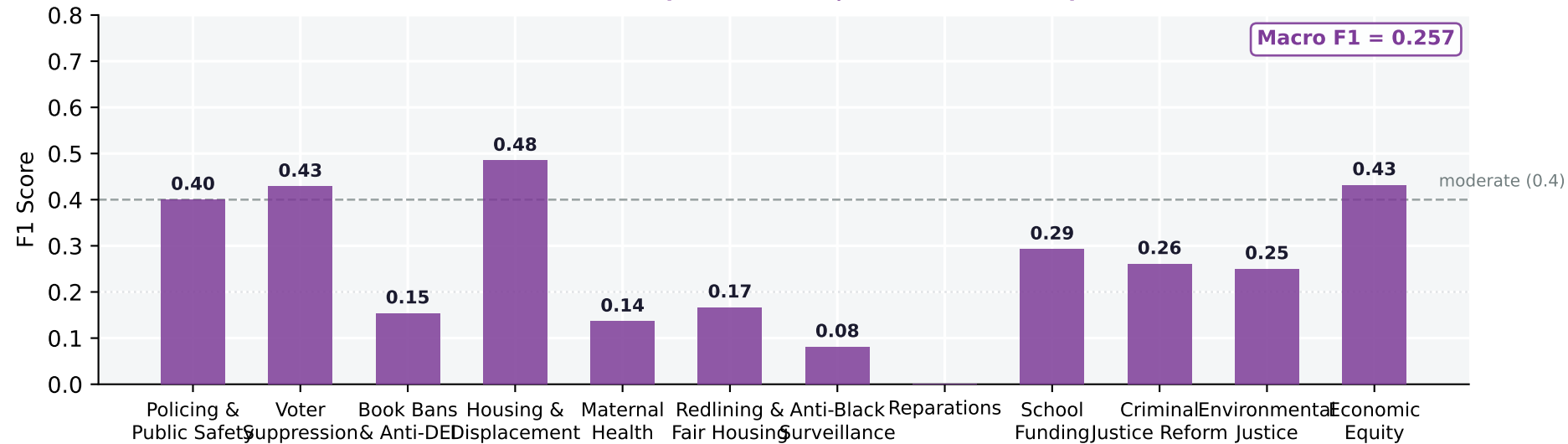


Each bar shows the F1 score for one topic — the harmonic mean of precision and recall.
A score above 0.4 (dashed line) indicates substantial agreement with the human annotator.
Topics with no bar (score = 0) were never tagged by the pipeline or never matched human labels.

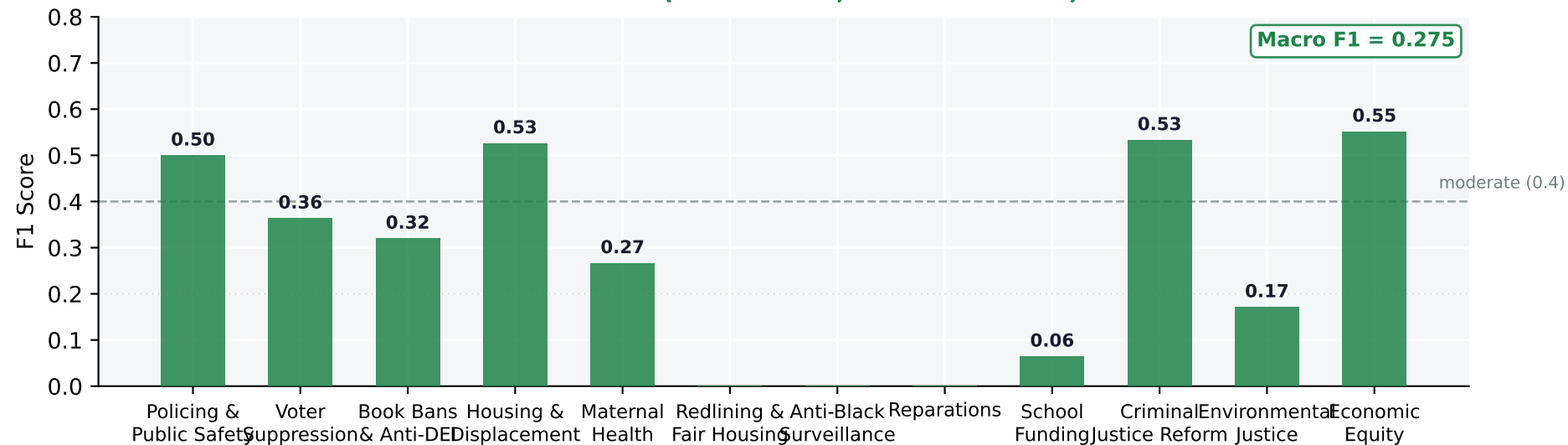
Semantic Tagger — F1 Score by Topic

How well does the semantic (AI embedding) tagger agree with each annotator on each topic?

Annotator 1 (200 articles, threshold=0.15)



Annotator 3 (200 articles, threshold=0.20)

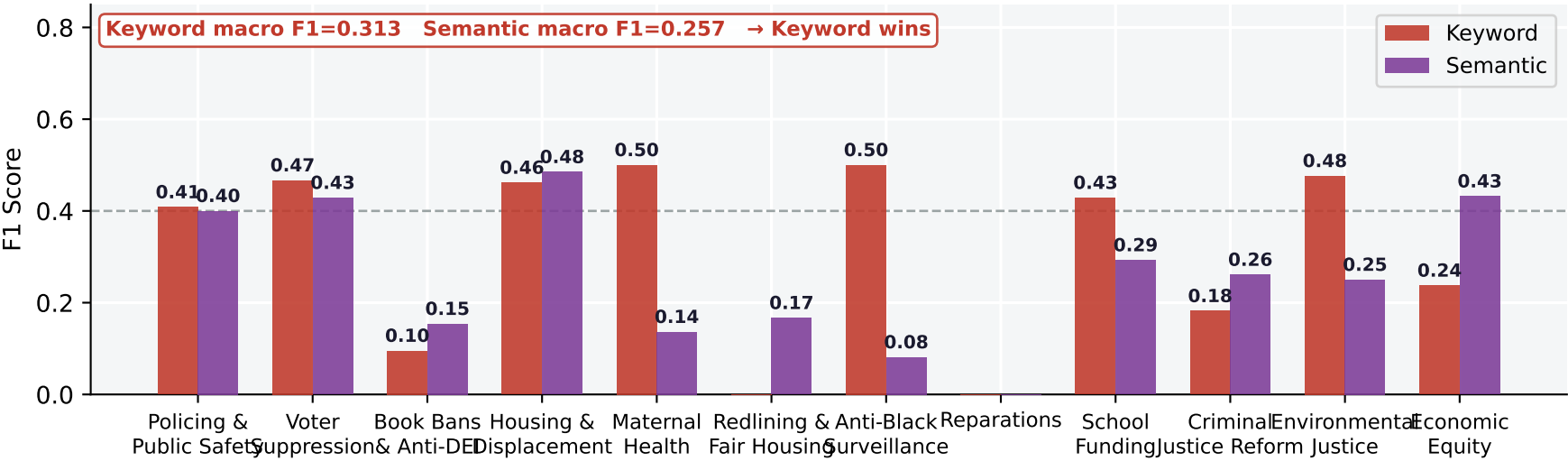


The semantic tagger converts each article and topic into an AI embedding vector, then tags articles whose similarity score exceeds the threshold (optimised per annotator). Threshold was chosen by scanning 0.10-0.35 and selecting the value that maximised macro F1.

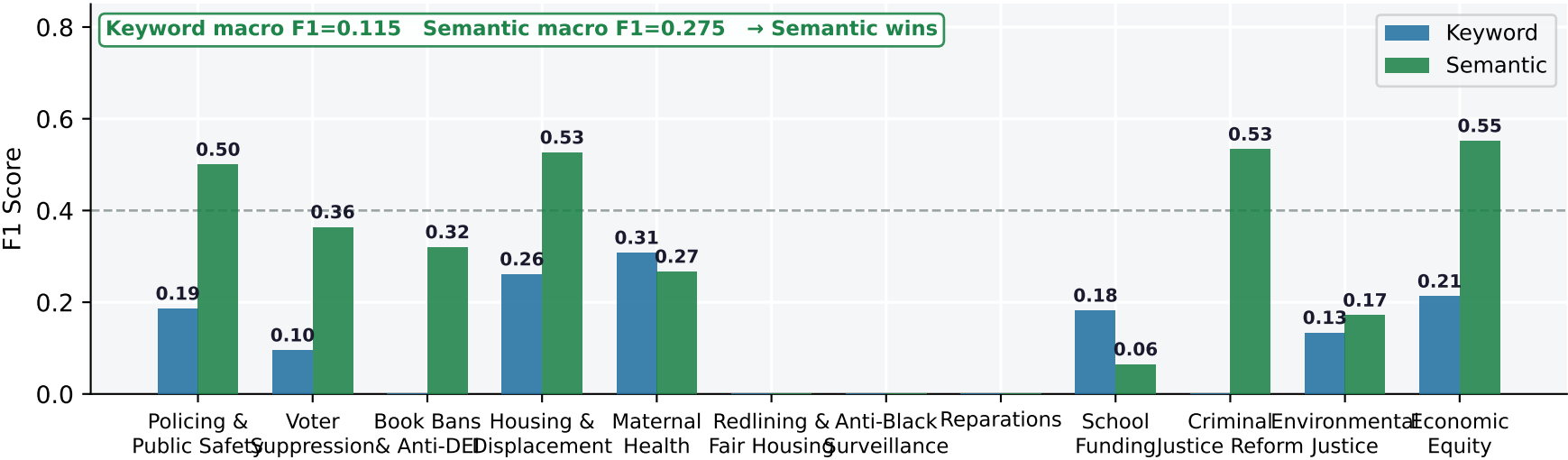
Keyword vs Semantic — Head-to-Head F1 by Topic

Direct comparison of both methods on the same article sets.

Annotator 1



Annotator 3



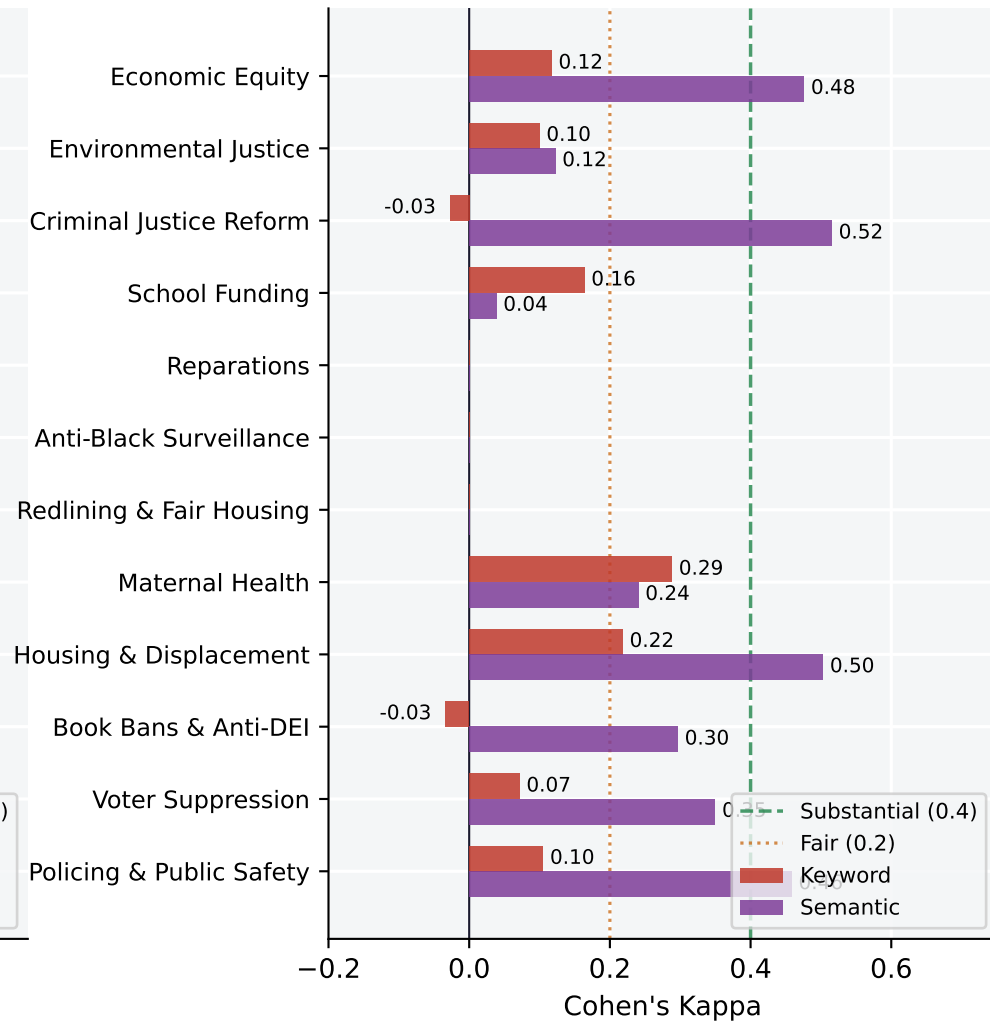
For each topic, the taller bar is the better-performing method on that annotator's article set.
The keyword pipeline outperforms on Annotator 1's set; the semantic tagger wins on Annotator 3's set.
This suggests neither method is universally superior — combining both may yield the best results.

Cohen's Kappa by Topic — Keyword vs Semantic

Agreement with human annotation corrected for chance. The most rigorous validation metric.

Annotator 1

Annotator 3



Substantial ($\kappa \geq 0.4$) Fair ($0.2 \leq \kappa < 0.4$) Poor ($\kappa < 0.2$)

Cohen's Kappa (κ) measures how much the pipeline and human annotator agree, beyond what would be expected by chance alone.

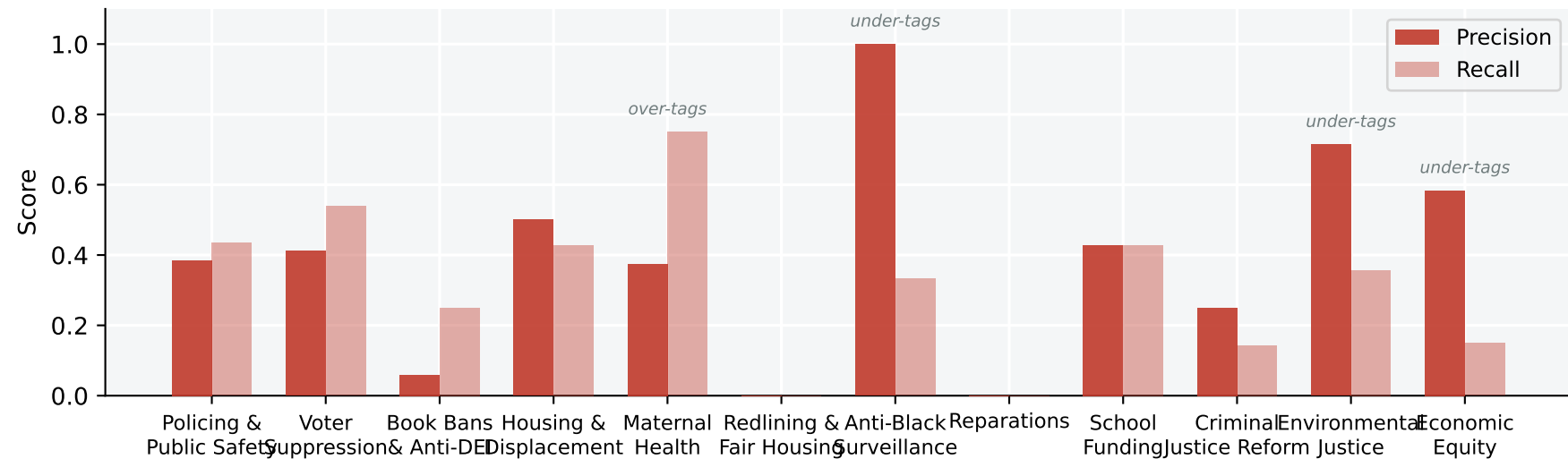
$\kappa \geq 0.4$ = substantial agreement (green dashed line) $\kappa 0.2-0.4$ = fair agreement $\kappa < 0.2$ = poor agreement.

Negative kappa means the pipeline disagrees with the human more than chance would predict.

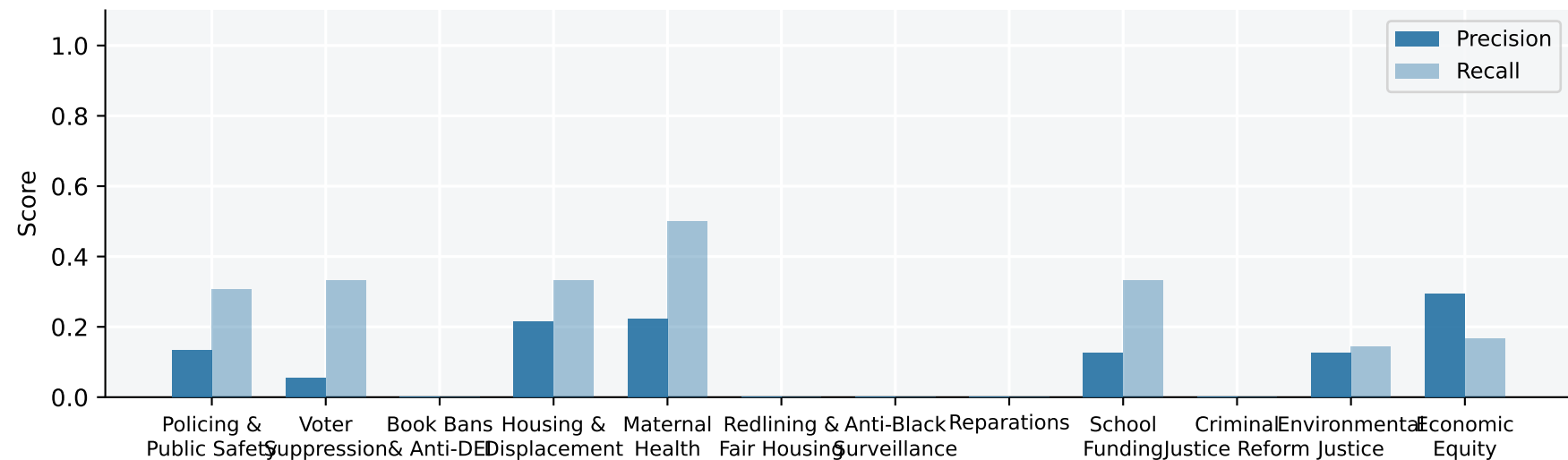
Keyword Pipeline — Precision vs Recall by Topic

Reveals whether the pipeline is over-tagging (too broad) or under-tagging (too narrow) per topic.

Annotator 1



Annotator 3



Precision (solid bar): when the pipeline assigns a topic, how often is the human in agreement?

Recall (faded bar): of all articles the human labeled for a topic, how many did the pipeline find?

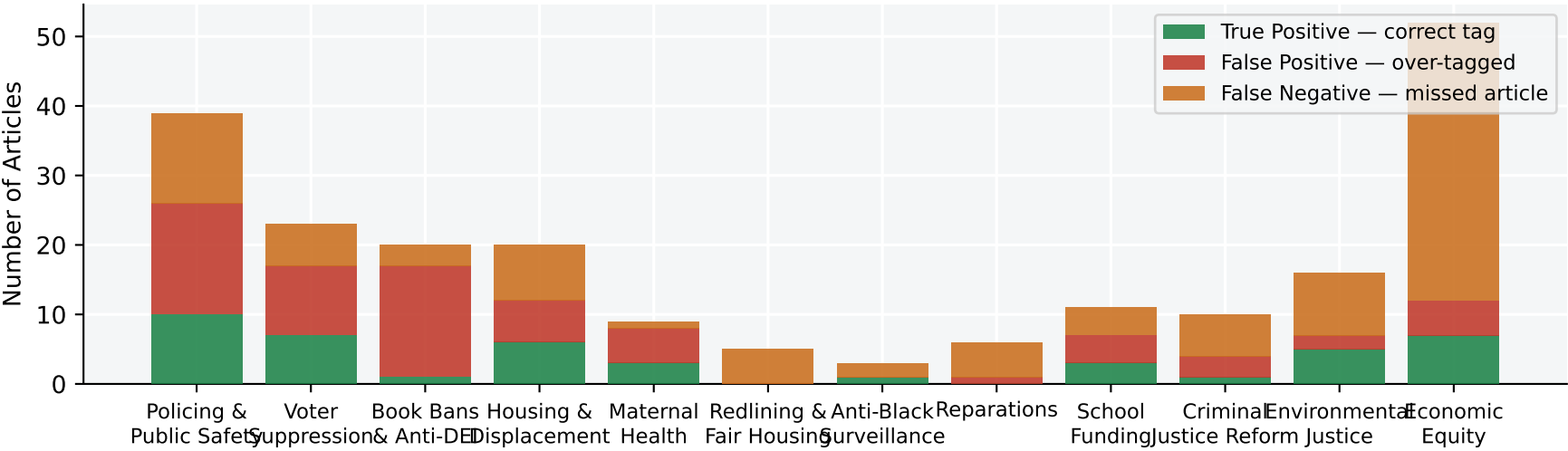
Over-tags → recall is much higher than precision (pipeline is too trigger-happy).

Under-tags → precision is much higher than recall (pipeline is too conservative, missing articles).

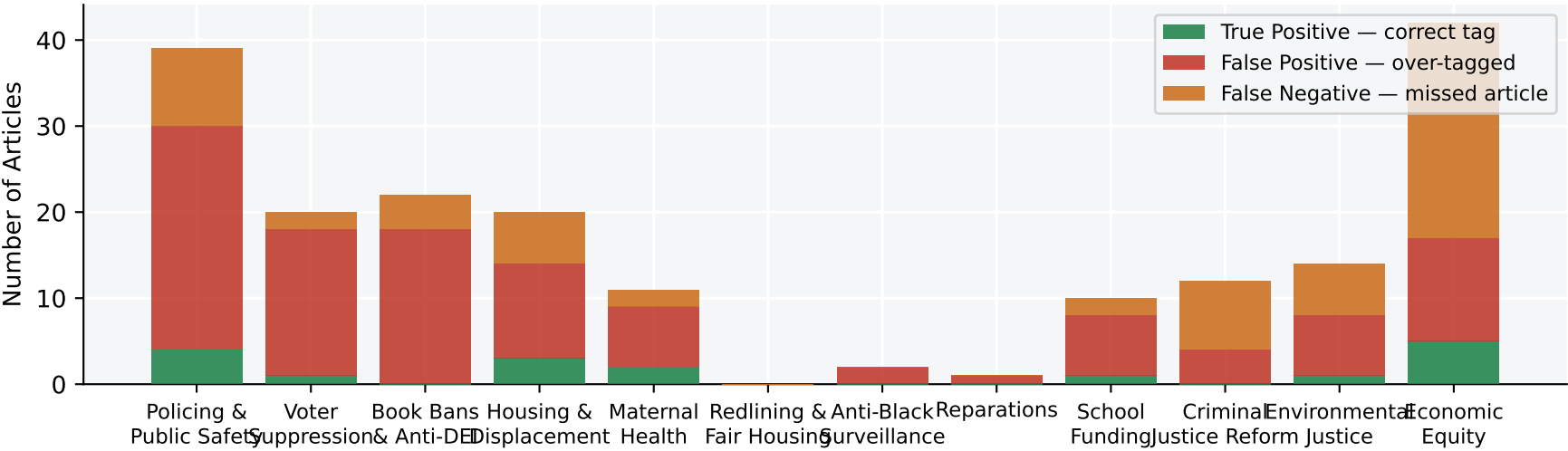
Keyword Pipeline — Error Breakdown by Topic

Exact counts of correct tags, false alarms, and missed articles per topic.

Annotator 1



Annotator 3

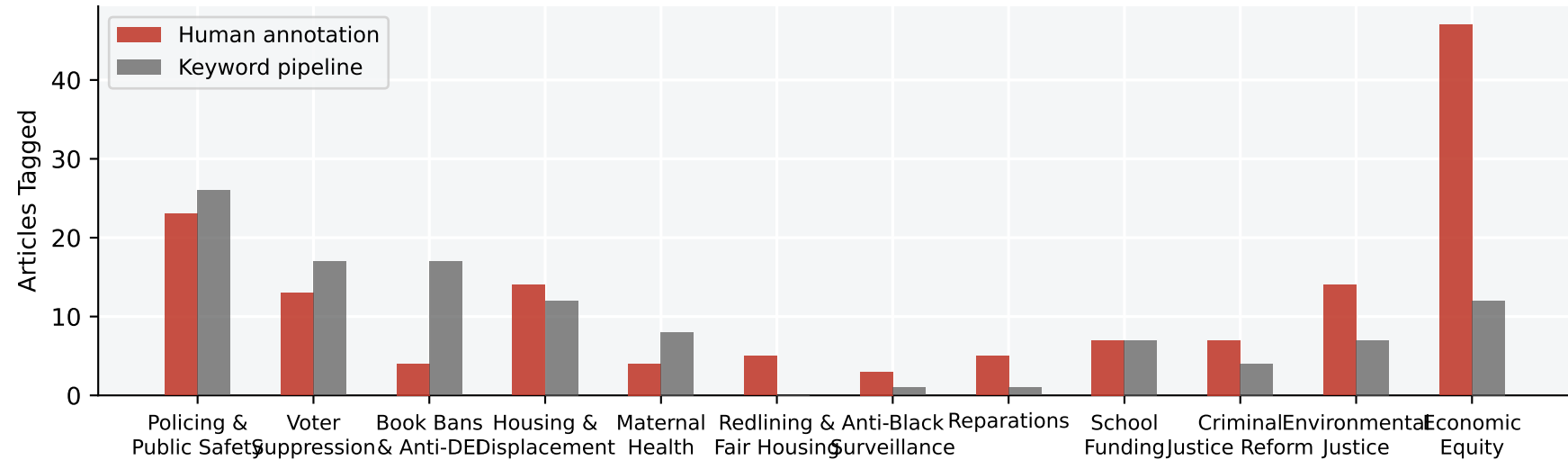


Green = both the pipeline and human agreed the topic is present (correct).
Red = pipeline tagged the topic, but the human did not (false alarm / over-tagging).
Orange = human labeled the topic, but the pipeline missed it (false negative / under-tagging).
An ideal topic has a tall green bar and minimal red and orange.

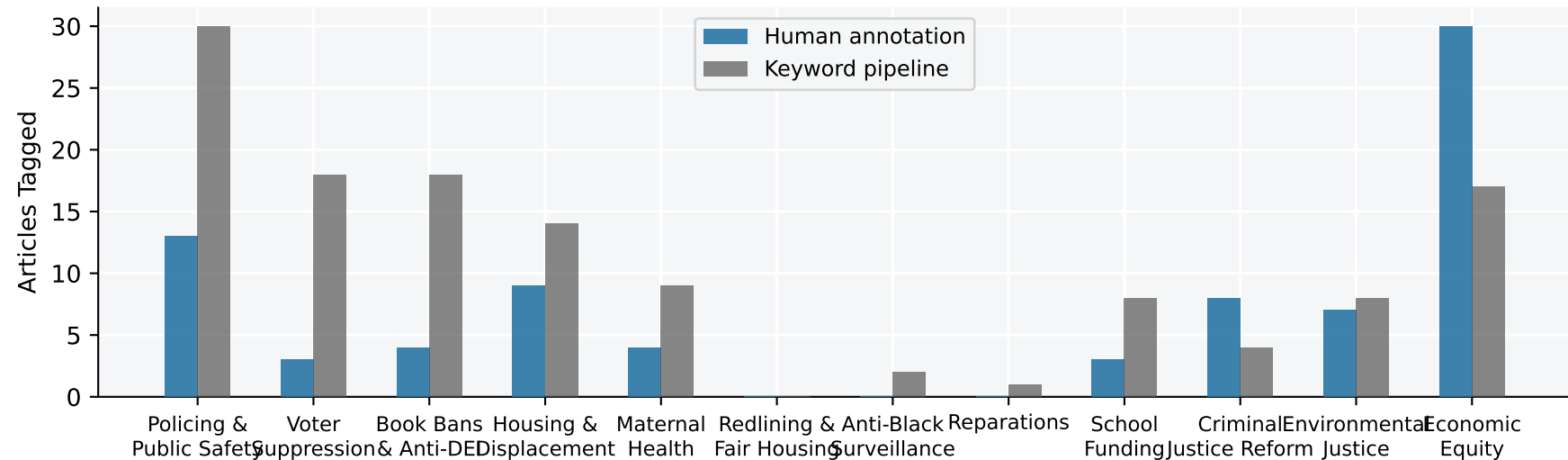
Articles Tagged per Topic — Pipeline vs Human

How many articles each method tagged for each topic vs how many the human annotator labeled.

Annotator 1 (200 articles)



Annotator 3 (200 articles)



This chart shows volume — how many articles were tagged per topic, not whether they were the right articles.

If the pipeline bar is much taller than the human bar, the pipeline is over-tagging.

If the pipeline bar is much shorter, the pipeline is under-tagging.

Equal heights do not guarantee accuracy — the pipeline may be tagging different articles than the human.